

Assuring Service Level Agreements in Open Radio Access Networks: An End-to-End System Design

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Abstract—We present a systematic design that ensures end-to-end service level agreements (SLAs) while minimizing the operational expenditure (OPEX) in an Open Radio Access Network (O-RAN). Unlike existing frameworks, which adopted theoretical models that are often misaligned with industrial standards, we devise our scheme based on standard-compliant, particularly 3GPP-specified models. We develop an algorithm that decouples the complex resource management into three coordinated closed-loop control stages: long-term infrastructure planning, medium-term computing resource allocation, and short-term radio resource scheduling. The algorithm has a short execution time, making it suitable for real-world implementations. We corroborate the efficacy of the proposed scheme through extensive numerical simulations and a prototype built upon an O-RAN testbed called INA-Infra, implemented using Nephio. Experimental results demonstrate that, compared to the baselines, our scheme significantly reduces SLA violations and enhances resource efficiency under non-stationary traffic conditions, confirming its engineering feasibility.

I. INTRODUCTION

Network Slicing is a key technology for 5G and next-generation communication networks to accommodate the various, and ever-increasing service demands [1]. In this paper, we demonstrate a translation of this concept from theory to practical deployment. Unlike most existing research, which is confined to isolated domains (either RAN or core), we present a holistic End-to-End (E2E) implementation, as well as optimization, using the INA-Infra prototype [2], a real-world experimental testbed based on O-RAN and Nephio.

A. Background

Although extensive research has addressed resource management in network slicing, transitioning these solutions to standard-compliant systems remains challenging due to three critical limitations discussed below.

Regarding the coordination of optimization timescales, [3] and [4] have proposed efficient algorithms for Virtual Network Function (VNF) placement and service chain mapping, whereas [5] further optimized CPU allocation in the core network to adapt to macroscopic traffic variations. In parallel, research focusing on the RAN domain, such as [6] and [7],

has developed dynamic scheduling frameworks to minimize latency or maximize spectral efficiency. However, these approaches often treat resource allocation in isolation across different domains. Specifically, the former focuses on static reservation while neglecting millisecond-level channel fluctuations, whereas the latter typically assumes fixed computing capacities, failing to coordinate with underlying infrastructure planning. This fragmentation across the compute-radio continuum limits the ability to handle E2E dynamics effectively.

In terms of mathematical modeling, representative works [8], [9] often rely on abstract resource models to construct utility functions and devise elegant resource allocation mechanisms. However, these theoretical models often lack precise mapping to industrial standards. For instance, in the 3GPP TS 28.541 standard, radio resource quotas are defined through complex attributes such as `RRMPolicyRatio`. Consequently, abstract models based on the (overly) simplified version of bandwidth cannot be directly translated into implementable configuration parameters for commercial O-RAN base stations (e.g., O-DU/O-CU), rendering theoretical gains difficult to reproduce in operational networks.

Finally, with the rise of network intelligence, [10] demonstrated the potential of Deep Learning for slice management, while [11] designed Deep Reinforcement Learning (DRL) specifically for latency-sensitive services. More recently, [12] proposed a DRL-based multi-timescale architecture. Nevertheless, despite improving automation, these methods struggle to provide the deterministic guarantees required for hard slicing. DRL-based agents operate as probabilistic “black-box” models that often require long online convergence times. Unlike optimization-based approaches, they cannot theoretically guarantee strict Hard Isolation or bounded Service Level Agreement (SLA) violations, posing significant risks for mission-critical communications.

B. Contributions

To address the above mentioned challenges, this paper proposes a standard-compliant end-to-end multi-timescale joint optimization algorithm. Unlike existing works that attempt to solve all dynamics in a monolithic model, we decouple the complex E2E slicing problem into three coordinated closed-loop control stages: *long-term* infrastructure planning, *medium-term* computing resource scaling, and *short-term* radio resource scheduling. This hierarchical algorithmic structure not only reduces computational complexity but also achieves a

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smooth transition from millisecond-level radio scheduling to second-level computing resource scaling via time-scale separation. Specifically, the proposed algorithm minimizes long-term resource reservation costs through infrastructure planning while dynamically adapting to traffic and channel fluctuations via medium-to-short-term loops to control dynamic allocation costs, thereby achieving an optimal balance between resource utilization and SLA assurance across the E2E link. Furthermore, we validated the proposed algorithm through extensive numerical simulations and a prototype implementation on INA-Infra [2] testbed. This comprehensive evaluation verifies both the engineering feasibility and the effectiveness of our algorithm.

The main contributions of this paper are summarized as follows:

- We establish an E2E network slicing resource allocation model strictly following 3GPP TS 28.541 [13]. By explicitly translating attributes like `RRMPolicyRatio` and `IsolationFlags` into optimization constraints, we bridge the gap between theoretical models and industrial standards.
- We propose a Three-Layer Multi-Timescale joint optimization algorithm. By coordinating long-term planning, medium-term scaling, and short-term scheduling, the proposed algorithm minimizes the operational expenditure (OPEX) while effectively guaranteeing SLAs.
- We validate the proposed algorithm through extensive numerical simulations and a prototype implementation on the INA-Infra testbed [2]. Experimental results demonstrate that, compared to static reservation or single-timescale schemes, the proposed algorithm significantly improves average achievable throughput and drastically reduces SLA violations. Furthermore, the hardware prototype confirms the engineering feasibility and validates the capability to effectively mitigate SLA violations and enhance system throughput in resource-constrained environments.

II. SYSTEM ARCHITECTURE

We develop our scheme based on the INA-Infra [2], an open, extensible, and O-RAN-compliant testbed that supports end-to-end system design. Below, we provide a brief introduction to this framework.

A. Hardware Components

The physical infrastructure of the testbed is organized into a hierarchical cloud topology that emulates a realistic telecommunications deployment environment. This topology comprises three distinct tiers: Edge Data Centers (Edge DCs), Regional Data Centers (Regional DCs), and Central Data Centers (Central DCs), as shown in Fig. 1. The Edge Cloud is deployed in close proximity to radio access points to provide low-latency access, albeit with limited computation and storage capacities. Conversely, the Central Cloud offers abundant resource pools suitable for delay-tolerant, large-scale

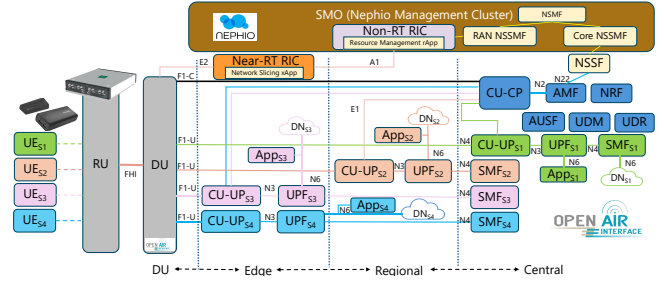


Fig. 1. System architecture of the hierarchical resource orchestration framework implemented on INA-Infra.

workloads, while the Regional Cloud provides a trade-off between resource capacity and transmission latency.

For the software stack, INA-Infra adopts a Kubernetes-based cloud-native foundation to manage distributed computing resources and integrates Nephio [14] as the core engine for automated orchestration. Nephio enables the platform to manage the lifecycle of network functions in a declarative manner through Custom Resource Definitions. The platform supports a full 5G protocol stack from OpenAirInterface (OAI) [15], allowing key network elements—including the Open Distributed Unit (O-DU), Open Central Unit (O-CU), and User Plane Function (UPF)—to be instantiated as containerized workloads, thereby achieving software-hardware decoupling and flexible deployment.

B. Control Units

The system employs a hierarchical control loop based on standardized O-RAN components. As illustrated in Fig. 1, the control logic is primarily distributed between the Non-Real-Time RAN Intelligent Controller (Non-RT RIC) within the Service Management and Orchestration (SMO) system and the Near-Real-Time RAN Intelligent Controller (Near-RT RIC) at the edge.

The Non-RT RIC is responsible for coarse-grained time-scale tasks and global resource optimization. Leveraging its global view of the network state, the Non-RT RIC hosts computationally intensive rApps to execute long-term topological planning and medium-term resource allocations. Specifically, it determines, on the one hand, the (optimal) physical deployment locations for network functions (e.g., CUs, UPFs) across Edge, Regional, and Central clouds based on long-term traffic statistics, and dispatches configurations via the O1 interface or the Nephio automation framework. On the other hand, it monitors network load changes using collected performance data and dynamically adjusts computation resource allocation quotas to adapt to fluctuations in traffic volume.

In contrast, the Near-RT RIC operates at the edge, near the RAN, and focuses on millisecond-level real-time control tasks. Through the standard E2 interface, the Near-RT RIC interacts directly with E2 nodes (e.g., DU, CU-UP) to acquire real-time radio channel quality indicators and scheduling requests. This component primarily hosts xApps (e.g., the Network-Slicing-xApp in Fig. 1) to execute fine-grained radio resource management strategies, such as dynamically allocating Phys-

ical Resource Blocks (PRBs) or enforcing isolation policies between slices.

This hierarchical design enables the system to use the Near-RT RIC to respond rapidly to transient channel conditions within the resource boundaries defined by the Non-RT RIC, thereby maximizing resource utilization while ensuring SLAs.

C. Design Objective

Given the hierarchical architecture and associated deployment costs, the primary objective of this paper is to minimize total OPEX while ensuring SLA compliance across different network slices. We concentrate on the User Plane functions (namely CU-UP and UPF) and the Application Server (APP), which dominate the resource consumption during the data transition path and directly affects end-to-end service delay. Specifically, the orchestrator is tasked with determining: (i) the deployment locations of the specific slice components (i.e., the CU-UP, UPF, and APP), (ii) the computing and memory resources allocated to each slice, and (iii) the radio resources scheduled for each slice. This requirement serves as the basis for the mathematical formulation detailed in the following section.

III. THEORETICAL FRAMEWORK

A. System Modeling

We denote by $\mathcal{S}$ the set of network slices. Each slice $s \in \mathcal{S}$ consists of a chain of virtual network functions, comprising the CU-UP, UPF, and APP, that need to be processed in sequence. The end-to-end SLA of slice s is characterized by a minimum throughput requirement $\bar{T}_s$ and a maximum end-to-end latency requirement $\bar{D}_s$. Moreover, to align with the isolation requirements defined in 3GPP, a slice can mandate hard isolation. We model this attribute using a binary indicator $H_s \in \{0, 1\}$, where $H_s = 1$ denotes hard isolation, requiring the minimum guaranteed resources to be fully backed by dedicated PRBs, while $H_s = 0$ indicates that such backing is not required.

We categorize the infrastructure nodes into three groups as $\mathcal{J} = \{\text{Edge, Regional, Central}\}$. Each element $j \in \mathcal{J}$ possesses certain computational resources. Specifically, for a generic node j , we denote by C_j^N and R_j^N the total CPU and RAM capacities available for hosting the CU-UP and UPF network functions. Similarly, for the APP services, we denote by C_j^A , G_j^A , and R_j^A the available CPU, GPU, and RAM capacities at node j , respectively.

In this system, the radio access segment comprises base stations located at the edge with PRB capacity B . To ensure that the theoretical framework aligns with real-world deployment, we leverage the key attributes defined in 3GPP to characterize the radio channels. Specifically, regarding the spectral resources for each slice s , we denote by $\hat{b}_s$ and $\tilde{b}_s$ the amount of dedicated PRBs reserved for and the minimum amount of PRBs allocated to this slice, respectively. Note that these two variables correspond to the `rRMPolicyDedicatedRatio` and `rRMPolicyMinRatio` attributes in `RRMPolicyRatio`

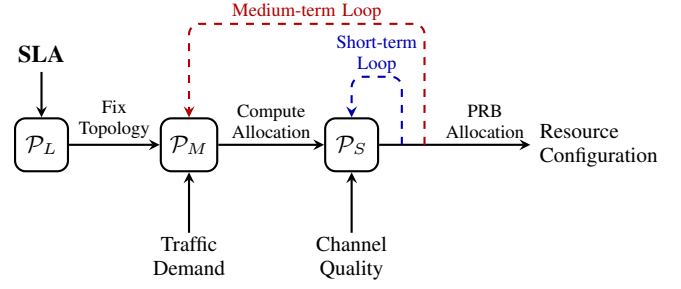


Fig. 2. Hierarchical logic flow: $\mathcal{P}_L$ enforces long-term topological deployment, $\mathcal{P}_M$ performs dynamic compute scaling, and $\mathcal{P}_S$ executes real-time PRB scheduling.

[13], respectively. We also model the isolation requirements from the slice configuration [13] as hard isolation constraints. When a slice s requires strict isolation (i.e., $H_s = 1$), the system enforces the constraint $\hat{b}_s = \tilde{b}_s$. This constraint ensures the minimum guaranteed allocation is strictly dedicated.

Regarding the radio transmissions, given the number of aggregated component carriers as I , 3GPP TS 38.306 [16] specifies the maximum data rate $\mathcal{R}$ (in Mbps) as follows:

$$\mathcal{R} = \frac{1.2 \times 10^{-5}}{T_s^\mu R_{max}^{-1}} \sum_{i=1}^I v^{(i)} Q_m^{(i)} f^{(i)} N_{PRB}^{BW^{(i)}, \mu} (1 - OH^{(i)}) \quad (1)$$

where $v^{(i)}$ denotes the maximum number of MIMO layers, $Q_m^{(i)}$ is the modulation order, $f^{(i)}$ represents the scaling factor, and R_{max} is the target coding rate. The term $N_{PRB}^{BW^{(i)}, \mu}$ denotes the maximum RB allocation in bandwidth $BW^{(i)}$ with numerology μ , T_s^μ is the average OFDM symbol duration, and $OH^{(i)}$ represents the overhead.

As the radio resource allocation varies across time slots, we normalize the maximum data rate onto the PRB basis by assigning $I = 1$ and $N_{PRB}^{BW^{(i)}, \mu} = 1$, which yields the expected throughput per PRB (namely, the spectral efficiency) at any typical time slot t :

$$\eta_s(t) = \frac{1.2 \times 10^{-5}}{T_s^\mu} \cdot v Q_m(t) R_{max}(t) (1 - OH) f \quad (2)$$

in which v and f are fixed system configurations, while $Q_m(t)$ and $R_{max}(t)$ are dynamic parameters determined by the real-time Modulation and Coding Scheme (MCS) index. We note that while the actual spectral efficiency may be nonlinear with the system parameters due to channel fading and interference, the linear approximation employed in our theoretical model establishes a conservative *safety margin* for slice provisioning. In practice, the scheduler can compensate for instantaneous fluctuations by leveraging the shared resource pool (i.e., resources not exclusively reserved), thereby ensuring the stability of the slice performance.

B. Problem Formulation

As the adjustment of resources residing on different parts of the network normally takes place on different time scales, we decompose the resource orchestration problem into three nested optimization layers: long-term planning, medium-term scaling, and short-term scheduling (cf. Fig. 2). The algorithm

follows a closed-loop, time-scale-aware design in which each layer operates at its appropriate granularity while exchanging constraints and feedback to ensure consistency in system behavior. Concretely, the long-term planning fixes the deployment topology, medium-term scaling adapts compute resources to traffic dynamics within this topology, and short-term scheduling allocates radio resources to meet instantaneous SLA requirements. Together, these layers provide a tractable and coherent framework for end-to-end resource orchestration.

1) *Long-term Planning Layer ($\mathcal{P}_L$)*: This layer determines the locations to deploy network functions and the corresponding resources to be reserved.

To establish an optimization problem that adequately accounts for the key features in this layer, we first define binary variables $x_{sj}, y_{sj}, z_{sj} \in \{0, 1\}$ that respectively indicate whether the CU-UP, UPF, and APP components of a slice s are deployed at node j . For the resources reserved across these components, we define continuous variables c_s^C and r_s^C to be the baseline compute and memory resources reserved for the CU-UP CPU and CU-UP RAM, respectively. Likewise, we define c_s^U and r_s^U as the resources reserved for the UPF CPU and UPF RAM, respectively. We also define variables c_s^A , r_s^A , and g_s^A as the CPU, RAM, and GPU reservations for APP services, respectively. Consequently, let P_j^C , P_j^R , and P_j^G denote the unit prices for CPU, RAM, and GPU at node j , respectively. We have the infrastructure cost, which aggregates the rental expenses for all allocated CPU, RAM, and GPU resources across the selected deployment nodes, given by

$$C_1(s) = \sum_{j \in \mathcal{J}} \left(P_j^C (c_s^C x_{sj} + c_s^U y_{sj} + c_s^A z_{sj}) + P_j^R (r_s^C x_{sj} + r_s^U y_{sj} + r_s^A z_{sj}) + P_j^G g_s^A z_{sj} \right). \quad (3)$$

We also define the radio cost of spectrum. Specifically, let P^{ded} and P^{prio} be the unit prices for dedicated and prioritized PRBs, respectively, the radio cost is

$$C_R(s) = P^{ded} \hat{b}_s + P^{prio} (\tilde{b}_s - \hat{b}_s). \quad (4)$$

For notational brevity, we aggregate the decision variables into deployment vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$, compute resource vectors $\mathbf{c}, \mathbf{r}, \mathbf{g}$, and radio resource vectors $\tilde{\mathbf{b}}, \hat{\mathbf{b}}$. Subsequently, we establish the following problem, aiming to minimize the infrastructure and radio costs while concurrently avoiding SLA violations, by optimizing a weighted sum of the cost and penalty:

$$\min_{\{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{c}, \mathbf{r}, \mathbf{g}, \tilde{\mathbf{b}}, \hat{\mathbf{b}}\}} Z_L = \sum_{s \in \mathcal{S}} \left(W^C (C_1(s) + C_R(s)) + W^P (\xi_s^D + \xi_s^R + \xi_s^C) \right) \quad (5)$$

in which W^C and W^P are weights associated with the cost and penalty, respectively, while the non-negative slack variables ξ_s^D , ξ_s^R , and ξ_s^C represent the degree of SLA violation in latency, radio resources, and compute resources, respectively.

$$\left\{ \begin{array}{ll} \sum_{j \in \mathcal{J}} x_{sj} = 1, \sum_{j \in \mathcal{J}} y_{sj} = 1, \sum_{j \in \mathcal{J}} z_{sj} = 1 & (6-1) \\ \sum_{s \in \mathcal{S}} (c_s^C x_{sj} + c_s^U y_{sj}) \leq C_j^N & (6-2) \\ \sum_{s \in \mathcal{S}} (r_s^C x_{sj} + r_s^U y_{sj}) \leq R_j^N & (6-3) \\ \sum_{s \in \mathcal{S}} c_s^A z_{sj} \leq C_j^A & (6-4) \\ \sum_{s \in \mathcal{S}} r_s^A z_{sj} \leq R_j^A & (6-5) \\ \sum_{s \in \mathcal{S}} g_s^A z_{sj} \leq G_j^A & (6-6) \\ r_s^C \geq MIN_R^{CU}, \quad r_s^U \geq MIN_R^{UPF} & (6-7) \\ T_s \leq \min\{\kappa_{CU} c_s^C, \kappa_{UPF} c_s^U, \gamma_C c_s^A, \gamma_R r_s^A, \gamma_G g_s^A\} & (6-8) \\ \xi_s^D \geq D_s - \bar{D}_s & (6-9) \\ \xi_s^C \geq \bar{T}_s - T_s & (6-10) \\ \xi_s^R \geq \bar{T}_s - \eta_s(t_0) \tilde{b}_s & (6-11) \\ 0 \leq \hat{b}_s \leq \tilde{b}_s, \quad \hat{b}_s \geq \tilde{b}_s \cdot H_s, \quad \sum_{s \in \mathcal{S}} \tilde{b}_s \leq B & (6-12) \\ x_{sj}, y_{sj}, z_{sj} \in \{0, 1\}, \quad \hat{b}_s, \tilde{b}_s \in \mathbb{Z}_{\geq 0} \\ c_s^C, c_s^U, c_s^A, r_s^C, r_s^U, r_s^A, g_s^A, T_s, \\ \xi_s^D, \xi_s^C, \xi_s^R \in \mathbb{R}_{\geq 0}, \quad \forall s \in \mathcal{S}, \forall j \in \mathcal{J} \end{array} \right.$$

This optimization problem is subject to the constraints in (6). First, (6-1) anchors each slice component to a unique location, while (6-2)–(6-6) enforce that resource allocation strictly respects physical node limits. (6-7) imposes a static RAM reservation, as the memory footprint of network functions is primarily dominated by fixed allocations rather than dynamic throughput variations. (6-8) models the achievable throughput T_s , where the coefficients κ and γ represent the resource efficiency factors. Notably, experimental observations confirm that packet processing exhibits a stable CPU cycle cost per bit under standard conditions [2], justifying this linear approximation where capacity is limited by the system bottleneck. (6-9)–(6-11) quantify SLA violations. Specifically, D_s in (6-9) denotes the end-to-end delay, calculated by aggregating the segments connecting the CU-UP, UPF, and APP:

$$D_s = \sum_{j \in \mathcal{J}} D_j^{F1} x_{sj} + \sum_{j, k \in \mathcal{J}} D_{jk}^{N3} x_{sj} y_{sk} + \sum_{k, l \in \mathcal{J}} D_{kl}^{N6} y_{sk} z_{sl} \quad (7)$$

where D_j^{F1} denotes the latency from the fixed DU to the CU-UP anchored at node j , D_{jk}^{N3} represents the transmission delay from the CU-UP at node j to the UPF at node k , and D_{kl}^{N6} corresponds to the delay from the UPF at node k to the APP at node l . For (6-11), t_0 denotes the planning decision time instant.

(6-12) enforces spectrum logic and mandates that for Hard Isolation, the minimum guaranteed throughput must be fully backed by dedicated PRBs.

$$\begin{cases} \sum_{s \in \mathcal{S}} (c_s^C(t)x_{sj}^* + c_s^U(t)y_{sj}^*) \leq C_j^N & (9-1) \end{cases}$$

$$\begin{cases} \sum_{s \in \mathcal{S}} (r_s^C(t)x_{sj}^* + r_s^U(t)y_{sj}^*) \leq R_j^N & (9-2) \end{cases}$$

$$\begin{cases} \sum_{s \in \mathcal{S}} c_s^A(t)z_{sj}^* \leq C_j^A & (9-3) \end{cases}$$

$$\begin{cases} \sum_{s \in \mathcal{S}} r_s^A(t)z_{sj}^* \leq R_j^A & (9-4) \end{cases}$$

$$\begin{cases} \sum_{s \in \mathcal{S}} g_s^A(t)z_{sj}^* \leq G_j^A & (9-5) \end{cases}$$

$$\begin{cases} T_s(t) \leq \min \{ \kappa_{CU} c_s^C(t), \kappa_{UPF} c_s^U(t), \\ \gamma_C c_s^A(t), \gamma_R r_s^A(t), \gamma_G g_s^A(t) \} & (9-6) \end{cases}$$

$$\begin{cases} \xi_s^{SLA-T}(t) \geq \bar{T}_s - T_s(t) & (9-7) \end{cases}$$

$$\begin{cases} \xi_s^{Demand-T}(t) \geq T_s^{demand}(t) - T_s(t) & (9-8) \end{cases}$$

$$\begin{cases} c_s^C(t), c_s^U(t), c_s^A(t), r_s^C(t), r_s^U(t), r_s^A(t), \\ g_s^A(t), T_s(t), \xi_s^{SLA-T}(t), \xi_s^{Demand-T}(t) \in \mathbb{R}_{\geq 0}, \\ \forall s \in \mathcal{S}, \forall j \in \mathcal{J} \end{cases}$$

2) *Medium-term Scaling Layer ($\mathcal{P}_M$)*: The Medium-term Scaling Layer operates on a second-level time scale to adapt to real-time traffic demand fluctuations. In this layer, the topological deployment decisions determined in $\mathcal{P}_L$, denoted as $\mathbf{x}^*, \mathbf{y}^*, \mathbf{z}^*$, are treated as fixed parameters. The decision variables include the dynamic resource allocation vectors $\mathbf{c}(t)$, $\mathbf{r}(t)$, and $\mathbf{g}(t)$, representing the CPU, RAM, and GPU resources allocated to the CU-UP, UPF, and APP components at time t , respectively.

The following problem minimizes the operational cost of active resources while balancing the trade-off between strict SLA compliance and dynamic demand tracking:

$$\begin{aligned} \min_{\{\mathbf{c}(t), \mathbf{r}(t), \mathbf{g}(t)\}} \mathcal{Z}_M(t) = & \sum_{s \in \mathcal{S}} \left(W^C C_I(s, t) \right. \\ & \left. + W^P \left(\xi_s^{SLA-T}(t) + \beta \cdot \xi_s^{Demand-T}(t) \right) \right) \end{aligned} \quad (8)$$

where $C_I(s, t)$ denotes the real-time infrastructure usage cost, calculated similarly to (3) but with dynamic resource variables, $\xi_s^{SLA-T}(t)$ and $\xi_s^{Demand-T}(t)$ represent penalties for SLA violations and demand mismatches, respectively, balanced by the weighting factor β .

This optimization problem is subject to the set of constraints formulated in (9). Specifically, (9-1)–(9-5) restrict the dynamic resource scaling strictly within the physical limits of the infrastructure nodes, anchored to the fixed deployment locations ($\mathbf{x}^*, \mathbf{y}^*, \mathbf{z}^*$). (9-6) applies the linear bottleneck model consistent with the planning layer, where the real-time throughput $T_s(t)$ is constrained by the minimum processing capacity among all allocated resources. (9-7) and (9-8) quantify service deviations, where (9-8) stipulates the system to align with the requested throughput $T_s^{demand}(t)$, defined as the summation of the packet arrival rate and the accumulated queue backlog from the previous cycle, ensuring that cloud resources are scaled to drain accumulated traffic and preventing VNF congestion.

$$\begin{cases} 0 \leq \hat{b}_s(t) \leq \tilde{b}_s(t) & (11-1) \end{cases}$$

$$\begin{cases} \sum_{s \in \mathcal{S}} \tilde{b}_s(t) \leq B & (11-2) \end{cases}$$

$$\begin{cases} \hat{b}_s(t) \geq \tilde{b}_s(t) \cdot H_s & (11-3) \end{cases}$$

$$\begin{cases} \xi_s^{PRB}(t) \geq \bar{T}_s - \bar{\eta}_s(t) \cdot \tilde{b}_s(t) & (11-4) \end{cases}$$

$$\begin{cases} \hat{b}_s(t), \tilde{b}_s(t) \in \mathbb{Z}_{\geq 0}, \quad \xi_s^{PRB}(t) \in \mathbb{R}_{\geq 0}, \quad \forall s \in \mathcal{S} \end{cases}$$

3) *Short-term Scheduling Layer ($\mathcal{P}_S$)*: The Short-term Scheduling Layer operates within the O-RAN Near-RT RIC on a timescale of 10 ms to 1 s. Unlike the underlying MAC scheduler, which handles millisecond-level channel fluctuations, this layer focuses on compensating for slow fading variations to satisfy slice SLAs. It optimizes the dynamic PRB allocation vectors $\hat{\mathbf{b}}(t)$ and $\tilde{\mathbf{b}}(t)$, i.e., the number of dedicated and minimum guaranteed PRBs, respectively.

We set up the following problem to minimize the radio costs while mitigating the risk of wireless SLA violations:

$$\min_{\{\hat{\mathbf{b}}(t), \tilde{\mathbf{b}}(t)\}} \mathcal{Z}_S(t) = \sum_{s \in \mathcal{S}} \left(W^C C_R(s, t) + W^P \xi_s^{PRB}(t) \right) \quad (10)$$

where $C_R(s, t)$ represents the radio resource cost, calculated using the pricing model given in (4) with dynamic variables $\hat{b}_s(t)$ and $\tilde{b}_s(t)$.

This optimization problem is subject to the set of constraints in (11). Particularly, (11-1) ensures that the PRB allocation adheres to 3GPP definitions (where dedicated resources are a subset of the minimum guaranteed), while (11-2) mandates that the aggregated allocation does not exceed the total cell bandwidth B . (11-3) enforces mission-critical requirements: if a slice is configured with the hard isolation flag ($H_s = 1$), its minimum guaranteed resources must be fully backed by dedicated PRBs. (11-4) quantifies potential service violations based on channel quality. To bridge the time-scale gap between millisecond-level channel dynamics and the $\mathcal{P}_S$ optimization cycle, we utilize the effective spectral efficiency $\bar{\eta}_s(t)$, obtained by temporally averaging the instantaneous efficiency $\eta_s(\tau)$ over the preceding $\mathcal{P}_S$ control cycle $[t - T_{PS}, t)$, where T_{PS} denotes the duration of one $\mathcal{P}_S$ cycle and N_{TTI} is the number of TTIs within this duration:

$$\bar{\eta}_s(t) = \frac{1}{N_{TTI}} \sum_{\tau \in [t - T_{PS}, t)} \eta_s(\tau) \quad (12)$$

This averaging process effectively filters out fast-fading fluctuations while retaining the slow-fading trends. Consequently, the penalty variable $\xi_s^{PRB}(t)$ captures the deficit between the SLA target $\bar{T}_s$ and the predicted throughput derived from this smoothed efficiency.

C. Complexity Analysis

To assess the algorithms' efficiencies, we analyze the worst-case complexity of solving the three decoupled sub-problems. First, we note that $\mathcal{P}_L$ can be reformulated from a Mixed-Integer Quadratic Programming (MIQP) problem to a Mixed-Integer Linear Programming (MILP) model by linearizing the

quadratic connectivity terms using standard auxiliary variables. The resulting (worst-case) complexity is determined by the Cartesian product of the primary binary topological variables and the integer radio resource variables, quantitatively, as $\mathcal{O}(2^{3|S||\mathcal{J}|} B^{2|S|})$. Given the long-term operational window (hours or days), this problem can be solved via standard branch-and-bound methods [17].

Second, $\mathcal{P}_M$ is a Linear Programming (LP) problem with a polynomial time complexity. With approximately 7 continuous decision variables per active slice, the complexity is $\mathcal{O}((7|S|)^{3.5})$. This polynomial complexity ensures that the system converges rapidly, making it suitable for adapting to traffic fluctuations within a seconds-level control loop.

Finally, $\mathcal{P}_S$ is an Integer Linear Programming (ILP) problem with a worst-case complexity $\mathcal{O}(B^{2|S|})$. While this implies the complexity might grow exponentially with the number of slices, practical tractability is ensured by two key structural factors: (i) the resource capacity constraints drastically prune the feasible search space from a high-dimensional hypercube to a smaller simplex, and (ii) the number of concurrent slices $|S|$ is generally small (typically 3 to 5 in commercial deployments), as network slices function as aggregated logical pipelines for service classes rather than dedicated tunnels for individual users.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

The simulation environment is set up based on the INA-Infra architecture, with parameters configured to reflect hardware heterogeneity (see Table I). Resources are distributed in a hierarchical manner: Edge DCs are resource-constrained with higher unit costs and lower latency, whereas Central DCs offer abundant, cost-effective capacity. Transport latencies are explicitly modeled based on the physical topology of each segment. First, the F1 interface latency corresponds to the CU-UP deployment location: 1 ms (Edge), 15 ms (Regional), and 30 ms (Central). Next, the N3 interface (CU-UP to UPF) introduces delays of 1 ms (Intra-site), 10 ms (1-Hop), and 20 ms (2-Hop) depending on the link span. Finally, the N6 interface (UPF to APP) imposes 5 ms, 15 ms, and 30 ms delays, likewise determined by the physical span of the connection.

TABLE I
HARDWARE CAPABILITIES AND RESOURCE COSTS

Variable	Unit	Edge DC	Regional DC	Central DC
C_j^N	ms	400	600	2500
C_j^A	ms	400	600	2500
R_j^N	GB	10	20	100
R_j^A	GB	20	40	200
G_j^A	Unit	150	200	1200
P_j^C	\$/100ms/h	0.5	0.05	0.001
P_j^R	\$/GB/h	0.1	0.01	0.002
P_j^G	\$/Unit/h	1.0	0.5	0.1

B. Scenario Definition

We evaluate a heterogeneous service scenario comprising four concurrent network slices ($\mathcal{S} = \{1, 2, 3, 4\}$), with diverse traffic profiles detailed in Table II. S1 represents background IoT traffic characterized by relaxed throughput and latency constraints. S2 models a delay-sensitive mission-critical control service. S3 simulates high-bandwidth HD video streaming, while S4 represents interactive applications (e.g., Cloud Gaming) requiring both high throughput and low latency.

The simulation discretizes the time domain, where a single time step corresponds to one scheduling cycle of the short-term layer ($\mathcal{P}_S$). Consequently, the medium-term layer ($\mathcal{P}_M$) executes resource scaling every 10 steps. To emulate time-varying wireless conditions, the MCS index is modeled as a stochastic process with temporal correlation. The MCS value is initialized via a uniform distribution and evolves as a Markov-like process, where the state at step t depends on step $t - 1$ subject to bounded random variations. This mechanism generates a dynamic spectral efficiency profile to examine the adaptability of our framework.

TABLE II
SIMULATION SCENARIO SLA REQUIREMENTS

Slice ID	$\bar{T}_s$ (Mbps)	$\bar{D}_s$ (ms)	H_s (Flag)
S1	10	100	0
S2	40	10	1
S3	100	35	0
S4	70	25	0

We evaluate the performance of the proposed algorithm against two baseline schemes. The static reservation baseline executes $\mathcal{P}_L$ only once at slice admission, keeping infrastructure and radio resource configurations fixed throughout the operational period. The single-timescale baseline executes $\mathcal{P}_L$ and synchronizes $\mathcal{P}_S$ with $\mathcal{P}_M$ to operate on the medium-term timescale, specifically every ten scheduling steps.

C. Simulation Results

First of all, we validate the efficacy of the proposed algorithm in a real-world deployment by measuring its execution time on standard commercial hardware (Intel Core i5-13400F CPU, 32 GB RAM) and comparing that with practical requirements. Our empirical results confirm that the optimization problems $\mathcal{P}_L$ and $\mathcal{P}_M$, defined for the long-term planning and medium-term control layers, respectively, can be solved in 68.4 ms and 1.1 ms, safely operating within their allowable time windows. Additionally, solving the more critical short-term scheduling problem $\mathcal{P}_S$ requires an average runtime of merely 1.8 ms. Given that the O-RAN Near-RT Control Loop is standardized to operate within a 10 ms to 1 s window, this performance confirms that the scheduler satisfies even the most stringent lower bound of the specification. As such, the proposed algorithm is proven computationally feasible and suitable for deployment within the O-RAN architecture.

Next, we examine the algorithm's decision-making capability at the long-term network planning layer. As detailed in

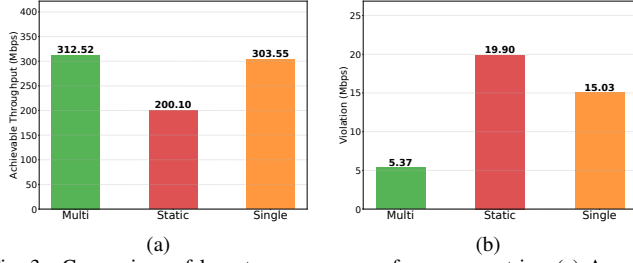


Fig. 3. Comparison of long-term average performance metrics. (a) Average achievable throughput. (b) Average violation throughput.

Table III, the algorithm formulates differentiated topological deployment strategies based on the SLA requirements of distinct slices: the latency-sensitive slice S2 is fully deployed at the Edge DC to ensure strict latency compliance, while the delay-tolerant slice S1 is consolidated at the Central DC to minimize costs. Conversely, for S3 and S4, the algorithm selects hybrid and regional topologies, respectively, to optimize the trade-off between performance and resource overhead. This strategic placement provides a robust baseline for the subsequent dynamic resource scheduling.

TABLE III
OPTIMAL COMPONENT PLACEMENT

Slice ID	CU-UP Location	UPF Location	APP Location
S1	Central	Central	Central
S2	Edge	Edge	Edge
S3	Regional	Central	Central
S4	Regional	Regional	Regional

Moreover, we investigate the system stability and efficiency under dynamic traffic conditions. We collect the statistics from a simulation spanning 10,000 time steps following full slice activation. The results are summarized in Figures 3 and 4.

Specifically, Fig. 3 presents a comparative analysis of the proposed method and the baselines on average performance metrics. As demonstrated by Fig. 3(a), the proposed algorithm attains the highest average throughput (of 312.52 Mbps), corresponding to 56.2% and 2.9% improvements over the static and single-timescale approaches, respectively. Meanwhile, Fig. 3(b) shows that the proposed method effectively contains the average throughput violation to 5.37 Mbps, marking a 73.0% and 64.3% reduction compared to the static and single-timescale baselines, respectively. These findings confirm that the proposed algorithm significantly enhances resource utilization while providing robust SLA assurance.

Fig. 4 illustrates the reliability performance by plotting the cumulative distribution function (CDF) of the SLA violation ratio, defined as the ratio of the violation amount to the minimum throughput requirement. The figure shows that the proposed algorithm achieves superior reliability with a non-violation rate (NVR), defined as the probability of zero SLA violation, of 85.88%, significantly surpassing the static (70.40%) and single-timescale (75.62%) baselines. Furthermore, the CDF of the proposed solution exhibits the steepest ascent, ensuring that the normalized violation severity remains negligible in the vast majority of cases. Conversely,

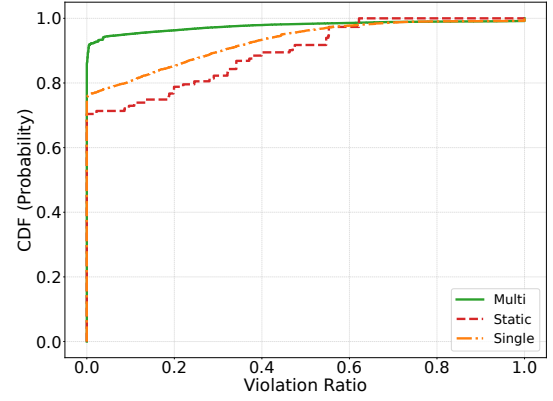


Fig. 4. CDF of SLA violation ratio.

the baselines exhibit distinct heavy-tailed behavior, indicating a higher risk of severe performance degradation in which a significant portion of the SLA remains unmet.

D. Validation on INA-Infra Testbed

In addition to simulations, we established an experimental testbed based on the INA-Infra architecture to validate the practical feasibility of our scheme. The platform hosts five heterogeneous slices with distinct SLA requirements (see Table IV). Slices S1 and S2 are deployed in Central and Regional nodes with relaxed constraints, while the time-critical slices S3, S4, and S5 compete for resources at the constrained Edge node. The current testbed represents a staged implementation focusing on RAN and Core dynamics, where the APP server is a lightweight UDP-based data server and does not yet leverage GPU acceleration. Specifically, we use the *iperf3* tool to emulate downlink traffic (e.g., video streaming) to stress-test the system's maximum throughput and validate SLA enforcement under resource contention.

TABLE IV
TESTBED EXPERIMENT SLA REQUIREMENTS

Slice ID	$\bar{T}_s$ (Mbps)	$\bar{D}_s$ (ms)
S1	25	100
S2	25	70
S3	30	50
S4	45	50
S5	90	50

The experimental timeline is divided into five distinct temporal regions (R1–R5) to simulate the progressive arrival of services. The throughput performance across these regions is illustrated in Fig. 5(a). In this figure, slices S1 to S4 are sequentially activated across regions R1 through R4. As observed, the proposed algorithm demonstrates robust dynamic adaptation. It effectively orchestrates resource allocation to satisfy the minimum throughput requirements of all active slices while utilizing residual capacity to maximize system utility. As a result, the system operates stably with no significant SLA violations during these intervals.

Region R5 represents the critical saturation scenario where all five slices are active. Specifically, slices S3, S4, and S5 compete simultaneously for the limited computing resources

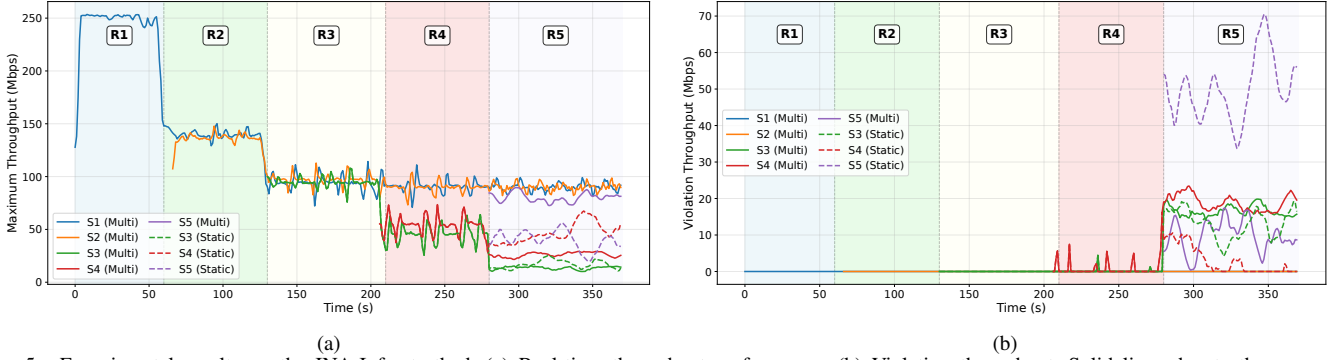


Fig. 5. Experimental results on the INA-Infra testbed. (a) Real-time throughput performance. (b) Violation throughput. Solid lines denote the proposed algorithm, while dashed lines represent the static baseline.

at the Edge node. To highlight the performance divergence under this bottleneck, we introduce a static baseline configured with fixed resource allocation via static deployment. Note that for clarity, this comparison focuses exclusively on the three Edge-hosted slices.

As aggregate demand in Region R5 exceeds the Edge node's physical capacity, SLA violations are unavoidable for both algorithms. However, the violation throughput analysis in Fig. 5(b) reveals a significant performance gap. The static allocation strategy lacks the flexibility to handle resource contention, leading to severe performance degradation, particularly for slice S5, which exhibits a high SLA violation rate. In contrast, the proposed algorithm actively rebalances the limited resources. Although it cannot create capacity out of thin air, it effectively mitigates congestion, resulting in substantially lower SLA violation rates and preserving the survivability of critical services under overload.

To quantify the performance benefits in this saturation scenario, we calculated the average metrics for the Edge-hosted slices in Region R5. The static baseline yields an average throughput of 103.23 Mbps but incurs a substantial average violation throughput of 66.03 Mbps. In contrast, the proposed algorithm improves the average throughput to 119.82 Mbps, representing a 16.1% gain in resource efficiency. More critically, it suppresses the average violation throughput to 45.18 Mbps, corresponding to a 31.6% reduction compared to the baseline. These results confirm that even under physical resource constraints, the proposed dynamic scheduling significantly enhances system capacity while mitigating the severity of SLA breaches.

V. CONCLUSION

We proposed a standard-compliant framework for end-to-end resource orchestration in O-RAN that bridges the gap between theoretical optimization and industrial implementation. Grounded in 3GPP-specified attributes rather than abstract models, our scheme ensures practical applicability. We developed a hierarchical control strategy that decouples resource management into coordinated long-, medium-, and short-term loops, effectively minimizing OPEX while strictly guaranteeing SLAs. Extensive simulations and prototype validation on the INA-Infra testbed confirm the framework's engineering

feasibility, demonstrating significant throughput and reliability enhancements under dynamic conditions.

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